

RICD Project Roadmap: Everything Deferred So Far

Compiled from every item put off for later across building, including ChatGPT's own suggestions and Carmen's own running to-do lists. Nothing here is started; this is a map of what's queued, not a commitment to order or timing. It does not include the four next trackers; while some of this would be completed while building those trackers, any other trackers will be built after the next four, to be built for: U.S. banking, hospitals, schools, and nursing homes.

1. The Controlled Mechanism Formalization Audit (ChatGPT's Proposal)

A structured, two-pass audit of every mechanism in the manuscript, designed specifically to stop the ghosted-formalization failure mode from recurring by making the epistemic status of every single mechanism explicit rather than assumed. This is a real, well-designed proposal worth doing as its own dedicated project, not folded into ordinary RICD version work.

Pass 1: Forensic Extraction

Go through the master manuscript mechanism by mechanism and extract, with no criticism, no rewriting, and no interpretation: the exact definition location, the equation as written, every variable it takes, and the stated conditions and consequences. This produces a clean, frozen baseline before any evaluation begins.

Pass 2: Adversarial Audit

Only after Pass 1 is frozen, ask three questions of every mechanism:

- A. Is the math actually executable? Could someone implement the equation from what's given, with no inferred assumptions? If not, mark MISSING and state specifically what's missing — not "probably obvious" or "the adapter handles it."
- B. Is the mechanism identifiable? Could a different, already-existing mechanism produce the same observed trajectory under current observables? If two mechanisms are observationally indistinguishable, state that explicitly ($M_1 \sim M_2$ under observables O) and identify what additional measurement would be needed to tell them apart.
- C. Is the adapter boundary explicit? For every input, classify it as RICD-derived, observed, fitted, adapter-supplied, or latent, so it's clear exactly where each quantity actually enters the system rather than left ambiguous.

The Per-Mechanism Audit Table

One row per mechanism, with these columns: Mechanism name, Definition location, Inputs, Domain/range, Trigger condition, Operator/equation, State transition, Parameter status (fixed/derived/model-derived/fitted), Source, Adapter inputs, Observables, Latents, Identification, Prediction, Falsifier, Status (structural/model-derived/empirical/computationally instantiated), and Citation.

Ablation Testing on the Higher-Ed Tracker

ChatGPT's other real suggestion from the same adversarial math-audit conversation, worth its own line rather than folding into the audit above: systematically remove pieces of the working tracker (M , G , R , Π_f , one at a time) and test alternative adapters, divergence functions, and functional forms, to see which components are actually carrying predictive signal. If the tracker keeps performing after removing half the machinery, or collapses after removing one

specific piece, either result is genuinely informative about which parts of the integrated framework matter empirically.

Dependency Graphs

For each mechanism, a small dependency chain (e.g., $D_{op}, D_{om}, D_{pm} \rightarrow I \rightarrow \Pi_f \rightarrow F_{boom} \rightarrow \dot{P}, \dot{R}$), with every arrow required to trace back to an actual equation or definition that justifies it. Any arrow that exists only because it seems like it should exist gets flagged rather than assumed.

The Mechanism Status Manifest

A second, standing manifest, separate from the Integration Manifest, tracking each mechanism against three separate yes/no/unclear columns: Formalized, Computable, and Identifiable — specifically to prevent the common error of treating "has an equation" as automatically meaning "is computable" or "is empirically testable," when those are three genuinely different claims.

The Strict Rules Governing This Audit

Given directly by ChatGPT, and worth preserving verbatim as the actual operating constraint for whoever runs this:

- Do not add mathematics. Do not repair mathematics. Do not infer missing formalizations.
- Do not declare something formalized because the manuscript discusses it conceptually — only record what is actually, formally present.
- Do not use memory of past conversations as evidence that something exists in the manuscript — the manuscript alone is the authority.
- Never silently fix anything found during the audit. A found issue is written as "FLAG: possible mathematical issue" and left for Carmen to decide on, not corrected in place and reported afterward as though it had always been that way.
- For every flagged item, present CLAUDE'S FLAG (what's wrong) and WHY (the specific textual gap), then let Carmen choose the resolution: add mathematics, add an empirical measurement, declare it explicitly unidentifiable under the current adapter, weaken the claim, or leave it as a conceptual mechanism rather than a fully testable one.

2. Documentation Projects

- RICD: A Concise Introduction — a standalone, roughly 20-page on-ramp document, distinct from the full manuscript, for a reader who needs the shape of the theory without the complete mechanism catalog.
- An abstract for RICD as a whole — a single, tight summary suitable for a paper submission, a grant application, or an investor deck, distinct from both the concise introduction and the full manuscript.
- A glossary of terms — every named mechanism, symbol, and piece of RICD-specific vocabulary in one alphabetized reference, so a new reader (or a co-author, or a reviewer) isn't stuck cross-referencing the full manuscript for every unfamiliar term.
- Full Status Taxonomy and Model-Status Table — ChatGPT's own single biggest-value recommendation from the very first adversarial review (item #6 of nine, RICD v5.9, August 20-21), deferred at the time specifically because it meant touching every row of an already-large set of tables. The actual project: classify every mathematical statement in the document into one of eight categories (primitive, axiom, definition, structurally derived consequence, model-derived default, fitted parameter, adapter choice, or explicit modeling assumption), then add that classification as a real status column across every row of Part 15's reference tables, so a reader can see at a glance whether a given quantity is forced by the axioms, one reasonable default among several, a number waiting to be fit, or left to the domain adapter's judgment. Part 15's tables are considerably larger now than when this was shelved.

- Constrained Function Class for the Multi-Valley Landscape Φ — the second of that same original pair, deferred because it depended on every forcing term acting on Φ landing coherently together, several of which were still being actively developed at the time. Rather than pinning down one exact formula for Φ 's shape, define the minimum properties any valid version must satisfy: the right number and position of local minima, rules for how barrier heights shift under forcing, the conditions under which a basin can disappear entirely, and continuity requirements tying the whole landscape together. Φ has had several more forcing terms and structural corrections added since (including the recent "three regions, not basins" softening), so this project's actual shape has likely changed since it was first proposed.
- A canonical, cross-cutting notation table — distinct from Part 15's existing eight reference tables, which are organized by originating Part rather than built to catch collisions across the whole document. One master table listing every symbol, its meaning, type, domain/codomain, and defining section, checked specifically for reused symbols, inconsistent subscripts, and definitions that drift between sections. Consolidation work on existing content, not new mathematics — most of what this catches has so far been found piecemeal through adversarial review (W_{ij} , γ_{net} , and others) rather than through one systematic pass.
- A structural validation test suite for the engine — a compact, non-empirical set of tests confirming the formal machinery behaves as specified, independent of any real tracker's data: probability distributions stay normalized, weights stay nonnegative, clipped quantities stay within their stated ranges, parent-ceiling ratios stay bounded, and known limiting cases (zero stress, maximal fragmentation, no memory, identical profiles, completely incompatible profiles) produce the behavior the manuscript claims. Tests the mathematics against itself, not against a domain.
- One complete, end-to-end worked example — a single case run through the full pipeline from a raw adapter input through RICD's core variables, FDFM dynamics, tracker output, and interpretation, with actual intermediate values shown at each stage plus one perturbation and its resulting change. Doesn't prove universal validity; demonstrates that the framework's definitions can actually be instantiated and calculated without hand-waving, in one place a new reader can follow start to finish.
- A consolidated known-limits section — Part 14 already states most of this piecemeal; gathering it into one explicit, standing list (which parameters are fitted rather than derived, which functional forms are model choices rather than theorems, which mechanisms remain conceptual rather than adapter-tested) would make the framework's own honesty about its own boundaries easier to find in one place rather than scattered across the document.

3. Prospective Trackers

Ten trackers planned beyond the completed higher-education one, plus the actor tracker as its own larger, separate project:

- Federal justice system tracker
- Medical-industrial complex tracker
- DV/rape-crisis funding-shift comparison tracker
- Forensic tracker: the 1929 crash, tied to the actor tracker's mechanism-layer approach
- Forensic tracker: the Titan submersible disaster, tied to the actor tracker's mechanism-layer approach
- Private equity / nursing homes tracker
- Scientific research integrity tracker
- Electrical grid resilience tracker
- Cybernetics / configuration-drift tracker — currently the strongest commercial candidate among the ten
- Private-equity / critical-infrastructure tracker (hospitals, utilities, housing) — surfaced in a Google AI conversation, not yet folded into the main list above but pinned as a real candidate. Flagged at the time: the proposed container hierarchy treated BlackRock and Vanguard as active "monopoly" control hubs directing cross-ownership, which conflates their real role as passive index-fund managers holding small percentages

across thousands of companies with a popular but factually shaky narrative — any future build needs a more defensible ownership-structure model before that specific framing is reused.

- The Collapse-Causal Agent Tracker itself — a distinct, larger, mechanism-layer project (actual agent behavior, not just downstream effects), deliberately not started yet per its own seed note, pending more collapse-causal trackers and real mechanism-layer case material (board minutes, depositions, whistleblower testimony, subpoenaed correspondence, OIG findings) rather than structured institutional data alone (for human agents; for many non-human agents, this tracker could be built now).

4. Academic and Institutional Applications

- Santa Fe Institute — Summer School application, and potentially a future presentation opportunity given how well RICD's actual profile (poetic/visual-spatial origin, adapter/engine architecture, real validated tracker results) appears suited to that specific audience.
- Antidisciplinary graduate programs: MIT Media Lab (MAS), NYU Tisch (ITP or similar), LIS, Arizona State University, Goldsmiths (University of London), UCL, and King's College London's LISS DTP.

5. Funding and Commercialization

- Emergent Ventures application — no submission deadline, nothing else needs to happen first.
- Angel investor applications/outreach — not yet started; would likely follow, not precede, having at least one more validated tracker or a working commercial prototype to point to.

The full monetization candidate list, grounded in real market research rather than priors, ranked by combined revenue potential and realistic data access:

Tier 1 — Best Combination of Lucrative and Buildable

- DevOps/SRE configuration-drift monitoring — confirmed real, large, fast-growing market (roughly \$4-12B depending on source, double-digit CAGR). Logs, metrics, and traces are the native, structured language of this industry, available in real time, with no privacy fights to win first. The strongest candidate overall — this is literally what the cybernetics tracker already targets.
- Manufacturing predictive maintenance — also confirmed large and fast-growing (estimates cluster around \$9-20B, some higher, consistently 15-30% CAGR). IoT sensor data is already flowing in most modern factories; this would build on existing telemetry rather than requiring new data collection. The collapse-onset-versus-confirmed-collapse distinction already built into RICD maps directly onto this.
- Universities/hospital administrator self-monitoring, repackaging the existing higher-ed tracker — smaller total addressable market than the two above, but the lowest-friction path to an actual first paying customer, since a working, validated model and public, structured data (IPEDS) already exist. Same underlying math, sold direct to leadership as an early-warning tool rather than built as an external public-interest tracker.

Tier 2 — Real Markets, Harder Data Access or Heavier Competition

- SaaS churn/product-health monitoring — large market, abundant proprietary usage data once under contract, but crowded with entrenched competitors (Gainsight, Amplitude, Mixpanel).
- Supply chain resilience monitoring — growing market, decent data availability, but fragmented across multiple providers, requiring real integration work before there's anything to sell.
- Corporate culture/HR monitoring — sizeable market, but dominated by well-funded incumbents (Culture Amp, Qualtrics, Glint/Google).
- Cybersecurity/insider-threat detection — big market, but one of the most heavily fortified, expertise-gated spaces in tech (CrowdStrike, Splunk); breaking in credibly would be its own multi-year project.

Tier 3 — Lucrative in Theory, Genuinely Hard to Access

- Commercial insurance underwriting — enormous market, but closed, actuarial, and deeply regulated; getting taken seriously as an outside data provider is a real barrier on its own.
- Hedge fund/institutional counterparty risk monitoring — high dollar value per client, but essentially requires an existing quant/finance reputation to get in the door.
- Franchise network health monitoring for franchisors — real, centralized data (franchisors already collect franchisee financials), but a niche market with a much smaller ceiling than the others.

Tier 4 — Real Need, Weakest Ability to Pay

- VC/PE portfolio company health dashboards — high-value clients, but data access is genuinely hard, since portfolio company financials are private and rarely standardized across a fund's holdings.
- Nonprofit/NGO board self-monitoring — public 990 data is real and available, but nonprofit budgets for internal tools like this are typically thin; the smallest realistic revenue ceiling on the list.

First-product scoping was already done for the top three Tier 1 candidates specifically, since they're the ones where the gap between an interesting idea and a sellable thing is smallest; that scoping work exists and can be picked back up directly rather than redone.

- Identifying further lucrative applications of RICD beyond the twelve above, as a standing, open-ended item rather than a single task with a defined endpoint.

6. Infrastructure and Maintenance

- GitHub repository: confirm the live IPEDS re-scoring pipeline is actually running successfully now, given the pipeline was still failing on live data fetches as of the last direct check.
- GitHub repository: keep the manuscript, the Integration Manifest, and the README's own described filenames synchronized with whatever the actual current RICD version is, each time a new version is finalized.